

Q19 — Lexical Analyzer

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
int main()
```

```
{
```

```
    char text[] =
```

```
        "#include <stdio.h>\n"
```

```
        "int main()\n"
```

```
        "{\n"
```

```
        "int number1, number2, sum;\n"
```

```
        "printf(\"Enter two integers: \");\n"
```

```
        "scanf(\"%d %d\", &number1, &number2);\n"
```

```
        "sum = number1 + number2;\n"
```

```
        "printf(\"%d + %d = %d\", number1, number2, sum);\n"
```

```
        "return 0;\n"
```

```
        "}\n";
```

```
    int i;
```

```
    int characters = 0;
```

```
    int words = 0;
```

```
    int lines = 0;
```

```
    int inWord = 0;
```

```
    for (i = 0; text[i] != '\0'; i++)
```

```
    {
```

```
        /* Count newline */
```

```
        if (text[i] == '\n')
```

```

        lines++;

/* Ignore spaces and tabs */
if (text[i] == ' ' || text[i] == '\t')
    continue;

/* Count words */
if (isalnum(text[i]) || text[i] == '_')
{
    if (inWord == 0)
    {
        words++;
        inWord = 1;
    }
}
else
{
    inWord = 0;
}

/* Count characters */
characters++;
}

printf("LEXICAL ANALYZER\n");
printf("-----\n");
printf("Number of Characters = %d\n", characters);
printf("Number of Words    = %d\n", words);

```

```
printf("Number of Lines    = %d\n", lines);

return 0;
}
```

Output

```
LEXICAL ANALYZER
-----
Number of Characters = 188
Number of Words      = 25
Number of Lines      = 10
-----
Process exited after 1.068 seconds with return value 0
Press any key to continue . . .
```